

Tian Bai

Postdoc | UiB

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► SUMMARY

I received my Ph.D. degree in computer science from the University of Electronic Science and Technology of China (UESTC) in June 2024 under the supervision of [Prof. Mingyu Xiao](#). My Ph.D. dissertation is entitled "[Investigations Concerning Parameterized and Exact Algorithms for the Feedback Set and Related Problems](#)". I am an EATCS member and CCF member, and my research interests include graph algorithms, approximation and parameterized algorithms, and algorithmic game theory.

► EMPLOYMENT

University of Bergen, Department of Informatics,
Postdoctoral Research Fellow

2025.09 - Present

- Host: Prof. Fedor Fomin.
- Research Interests: Parameterized Complexity, Clustering, Mechanism and Contract Design.

The University of Hong Kong, Department of Computer Science,
Postdoctoral Research Fellow

2024.09 - 2025.08

- Host: Prof. Zhiyi Huang.
- Research Interests: Parameterized Algorithms, Auction Theory, Optimal Stopping.

► EDUCATION

University of Electronic Science and Technology of China, School of Computer Science and Engineering,
Computer Science, *Ph.D.*

2024.06

- Advisor: Prof. Mingyu Xiao.
- Research Interests: Graph Algorithms, Parameterized Algorithms, Mechanism Design.

University of Electronic Science and Technology of China, BS Class (Math & Physics),
Fundamental Sciences (Mathematics and Physics), *B.Sc*

2016.06

- Research Interests: Algebraic Coding Theory, Information Theory.

► RESEARCH PUBLICATIONS

Journal Articles

- [1] **Tian Bai**, Mingyu Xiao: [Solving Subset Feedback Vertex Set in Chordal Graphs Faster than \$2^k\$](#) . Information and Computation (Accepted)
- [2] **Tian Bai**, Mingyu Xiao: [A Parameterized Algorithm for Subset Feedback Vertex Set in Tournaments](#). Theoretical Computer Science 975: 114139 (2023).
- [3] **Tian Bai**, Mingyu Xiao: [Exact Algorithms for Restricted Subset Feedback Vertex Set in Chordal and Split Graphs](#). Theoretical Computer Science 984: 114326 (2024).
- [4] **Tian Bai**, Mingyu Xiao: [Computational Complexity of Feedback Vertex Set and Subset Feedback Set Problems: A Survey](#). Journal of computer research and development (2024).
- [5] **Tian Bai**, Mingyu Xiao: Exact Algorithms for Maximum Independent Set Problem on Hypergraphs. SCIENTIA SINICA Informationis.

Conference Proceedings

- [1] **Tian Bai**, Zhiyi Huang, Chui Shan Lee, and Dongchen Li.: [Optimal Stopping with a Predicted Prior](#). Proceedings of the 58th ACM Symposium on Theory of Computing (EC 2026)
- [2] **Tian Bai**, Yiding Feng, Yaohao Liu, Mengfan Ma, and Mingyu Xiao: [Contracting with a Mechanism Designer](#). Proceedings of the 27th ACM Conference on Economics and Computation (EC 2026).
- [3] **Tian Bai**, Yixin Cao, Mingyu Xi: [Feedback Set Problems on \(Planar\) Graphs of Bounded Degrees](#). Proceedings of the 32nd International Computing and Combinatorics Conference (COCOON 2026).
- [4] Yingying Huangfu, **Tian Bai**: [On the Leaky Private Information Retrieval with Side Information](#). IEEE Information Theory Workshop, (ITW 2026).
- [5] **Tian Bai**, Fedor Fomin, Petr Golovach, Yash Hiren More, Simon Wietheger Clustering Permutations under the Ulam Metric: A Parameterized Complexity Study Proceedings of the 53rd International Colloquium on Automata, Languages and Programming (ICALP 2026).
- [6] Junqiang Peng, **Tian Bai**, Jingyang Zhao, Mingyu Xiao: [Sustained Vertex Cover on Temporal Graphs](#). Proceedings of the 35th ACM on Web Conference (WWW 2026).

[7] Yingying Huangfu, **Tian Bai**: [A Cooperative Statistical Approach for Abnormal Node Detection with Adversary Resistance](#). Proceedings of the 34th International Conference on Computer Communications and Networks (ICCCN 2025).

[8] **Tian Bai**, Mingyu Xiao: [Breaking the Barrier \$2^k\$ for Subset Feedback Vertex Set in Chordal Graphs](#). Proceedings of the 49th International Symposium on Mathematical Foundations of Computer Science (MFCS 2024).

[9] Mengfan Ma, **Tian Bai**, Mingyu Xiao, Xin Chen: [Facility Assignment with Fair Cost Sharing: Equilibrium and Mechanism Design](#). Proceedings of the 30th International Computing and Combinatorics Conference (COCOON 2024).

[10] Mengfan Ma, Mingyu Xiao, **Tian Bai**, Bakh Khoussainov: [Facility Location with Entrance Fees](#). Association for the Advancement of Artificial Intelligence (AAAI 2023).

[11] **Tian Bai**, Mingyu Xiao: [Exact and Parameterized Algorithms for Restricted Subset Feedback Vertex Set in Chordal Graphs](#). Proceedings of the 17th Annual Conference on Theory and Applications of Models of Computation (TAMC 2022).

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SERVICES AND PROFESSIONAL ACTIVITIES

Reviewer for Conferences

- ICALP 2026
- ESA 2026
- ISAAC 2024
- MFCS 2024
- IPEC 2025
- COCOON 2025
- SOFSEM 2025
- WALCOM 2022 2024 2025 2026.

Reviewer for Journals

- Theoretical Computer Science.
- Journal of Computer and System Sciences.
- Discrete Applied Mathematics.
- Frontiers of Computer Science.

Moderator for Workshops

- The 19th International Conference and Workshops on Algorithms and Computation (WALCOM), 2025.
- CCF National Conference of Computational Economics, 2024.
- Algorithms & Logic Workshop at UESTC, 2023.
- Algorithmic Game Theory and Computation Theory Forum at UESTC, 2022.
- The 2nd Theoretical Computer Science Outstanding PhD Students Forum (TCSPhd2021), 2021.

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TEACHING

Teaching Assistant for graduate students: Frontier Topics in Doctoral Research	2022-2023
Teaching Assistant for graduate students: Advanced Algorithm Analysis and Design	2022-2023
Teaching Assistant for undergraduate students: Frontiers of Theory of Computation	2022-2023
Teaching Assistant for undergraduate students: Frontiers of Theory of Computation	2021-2022
Teaching Assistant for undergraduate students: Advanced Algebra	2017-2018
Teaching Assistant for undergraduate students: Linear Algebra	2016-2017

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AWARDS AND HONORS

Outstanding Teaching Assistant	2024
Global 69th/China 10th place in IEEE Global Extreme Programming Competition 12.0	2018
First prize in the National University Mathematics Competition	2014 & 2015
Scholarship: People’s First Class Academic Scholarship (multiple times)	2012 – 2015